

Aankit Das

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Evidence of Excellence

- Authored 4 peer-reviewed papers with 140+ citations
- Awarded Jonsson School Dean's Scholarship (twice) for research excellence at UT Dallas.

Technical Skills

Python, C, C++, MATLAB, Bash, SQL, SystemVerilog, VHDL, PyTorch, Transformers, NLP, LLMs, RAG, Scikit-learn, Distributed Training, CUDA, cuDNN, Hydra, PySM3, TinyML, TensorFlow, Keras, Scikit-learn, Flask, Docker, AWS, MS SQL Server, Alteryx, Tableau, GEM5, Git, Docker, Shell, CMake, PowerShell

Education

University of Texas at Dallas

Aug 2023 - May 2025

MS in Computer Engineering, GPA: 3.8/4.0

Coursework: Machine Learning, Statistical Methods in AI and ML, NLP, Applied Data Structures and Algorithms, Computer Architecture, Embedded Systems, Wireless Sensor Systems, Hardware Modelling, Advanced Digital Language

University of Calcutta

Aug 2016 - Dec 2020

BTech in Electronics and Communication Engineering, GPA: 3.7/4.0

Thesis: Low-Power Multi-Optional Sat-nav Module with Multi-Constellation Satellite Signal Tracking. [Link](#) 

Experience

AI Engineer

Dallas, TX

Bantrly

March 2026 - Present

- Designed and built a modular speech-to-text routing engine (FastAPI, Python) that separated transcription from LLM scoring across 8 K-12 speech assessment modules, cutting audio API costs and improving scoring rubric.
- Implemented a two-layer VAD chunking pipeline — silence-boundary audio chunking with timestamp deduplication, plus a sliding text window for guardrail checks — to eliminate boundary-induced false positives during real-time streaming.
- Built a provider-agnostic adapter registry (Parakeet TDT, Whisper, AssemblyAI, Gemini) with quality-based fallback and session health checks, driven entirely by config so engines swap with zero code changes.
- Evaluated neural TTS engines (pyttssx3, Kokoro, edge-tts, Chatterbox, Chirp) against a structured rubric — intelligibility, naturalness, pacing, latency — per K-12 grade band, producing cost-quality comparisons to guide engine selection.
- Designed an LLM-as-judge evaluation pipeline with structured JSON output, retry logic, and cosine-similarity consistency scoring via nomic-embed-text embeddings, running 468 controlled experiments across 3 open-source models to quantify quality, consistency, and hallucination trade-offs.

Machine Learning Researcher

Richardson, TX

University of Texas at Dallas

June 2025 - March 2026

- Building and optimizing neural pipelines within [ADAPT RAISE](#) , an **NSF-funded** effort (Award #2327245) to accelerate cosmological inference using deep learning, focusing on extracting clean power spectra from noisy CMB (Cosmic Microwave Background) maps, enabling faster, scalable alternatives to traditional Bayesian pipelines.
- Active contributor to the [CMB-ML GitHub organization](#) , supporting open-source research for cosmology and machine learning; working toward submitting findings to ICCV/CVPR.
- Processed and managed 1,600 high-resolution (Nside=512) CMB simulated maps (24GB) across multiple splits.
- Developed an **UNet-inspired** encoder-decoder neural architecture, trained under 20 seconds per epoch on modern GPUs, for denoising and recovering the angular power spectrum (C_ℓ) across $\ell \in [2, 1280]$ from these simulated maps.
- Performed rigorous **ablation studies** including dropout, batch normalization, loss function variants (MSE, normalized MSE, Wishart), and activation functions (ReLU, GELU).
- Automated hyperparameter sweeps using Hydra and shell scripting to benchmark model performance across ℓ s, detectors, and feature configurations at different resolutions (Nside=128/512).

Data Scientist

Kolkata, WB

SEI Investments Company

Oct 2021 - Aug 2023

- Developed a **Python-based** prototype system for automating document-filling processes using **NLP** libraries and **OCR** technology to reduce manual processes by 50%, enhancing data availability.

- Architected a comprehensive **reconciliation workflow** for diverse investor profiles using **Alteryx**, boosting efficiency and ensuring seamless integration with cross-functional teams.
- Implemented error-handling mechanisms to enhance reliability and documented the **algorithm design**, implementation steps, and **performance metrics** for knowledge sharing within the team.
- Built **Tableau** dashboards for large-scale datasets, offering actionable insights for business stakeholders.

Research Internships

Research Assistant

University of Nebraska-Lincoln

Remote

May 2021 - Jun 2023

- Collaborated with [international teams](#) to develop **EmergeNet**, a novel segmentation model achieving 99.4% IoU (in comparison, the standard UNet has an IoU of 69%) for agricultural data.
- Developed the University of Nebraska-Lincoln Maize Emergence Dataset ([UNL-MED](#)), a **benchmark dataset** featuring top-view time-lapse images of maize coleoptiles from pre-emergence to approximately one inch in height, used for evaluating the performance of EmergeNet.
- Applied **optimization** techniques to monitor the growth of maize coleoptiles with a correlation factor of 0.99 between predicted and observed growth metrics, publishing findings in [Frontiers](#) and an [e-book](#).

Research Assistant

Jadavpur University

Kolkata, WB

Apr 2020 - Aug 2021

- Innovated nature-inspired meta-heuristic algorithms for feature selection ([BBA-LAHC](#) & [HS-NMR](#)), achieving over **98%** test accuracy in language identification.
- Published two **IEEE journal papers** showcasing advancements in NLP and audio signal processing.

Projects

Bantrly Lesson Generator — K–12 ELA Content Generation System

[Demo](#) [GitHub](#)

- Built a **hybrid LLM + rule-based lesson generation system** for K–12 ELA speaking practice — combining Groq inference, a CCSS-aligned 80-skill taxonomy, few-shot RAG, and a 9-step validated pipeline to produce structured, grade-appropriate lessons from three inputs.
- Designed and implemented **multi-layer guardrails**: pre-generation input validation, post-generation content checks (vocabulary ceiling, cognitive load, single-skill enforcement), and a **cultural bias retry loop** — flagged terms injected back into the prompt as explicit warnings across up to 5 retries before raising an error; no biased lesson is ever saved.
- Engineered a **defensive validation pipeline** handling non-deterministic LLM output: auto-correction for three distinct practice field structures across model variants, reflect stage normalization, and Speaking-domain field stripping — ensuring clean JSON regardless of model behaviour.
- Built a **CCSS-aligned skill taxonomy** (80 skills across 4 grade bands × 4 domains) with deterministic coverage tracking and cycling — replacing free-form LLM skill generation with controlled, measurable, progression-aware objectives.
- Deployed a **6-tab Gradio UI** on Hugging Face Spaces with a student-facing step-by-step lesson experience, coverage heatmap (Plotly), skill taxonomy browser (responsive CSS grid), and guardrail inspector — all updating live after each generation.
- *Tools: Python, Groq (llama-3.3-70b / llama-3.1-8b), Pydantic, Gradio, Plotly, GitHub Actions, Hugging Face Spaces, uv*

Document Intelligence Platform — Production RAG System

[Demo](#) [GitHub](#)

- Engineered production-grade Retrieval-Augmented Generation system combining local embedding models (Ollama, Sentence Transformers), vector storage (Chroma DB), and LLM APIs (Groq llama-3.1-8b-instant) for intelligent document question-answering.
- Designed modular Python architecture (5 independent modules: chunker, embeddings, vector_store, llm, pipeline) enabling component swapping, testability, and maintainable codebases aligned with software engineering best practices.
- Implemented FastAPI REST endpoints with async request handling, Pydantic type safety, health checks, and structured logging for production observability.
- Built semantic chunking pipeline (500-token chunks, 50-token overlap) with PyPDF2 extraction, generating 768-dimensional embeddings for cosine similarity retrieval with less than 10ms latency on 100+ document chunks.
- Deployed containerized application using Docker multi-stage builds, automated CI/CD via GitHub Actions, and hosted on Hugging Face Spaces with serverless scaling.
- Demonstrated cost-optimization strategy: local embeddings for privacy and efficiency, free-tier LLM APIs for fast inference, persistent vector storage for retrieval performance.
- *Tools Used: PyTorch, Transformers, Python, FastAPI, Docker, GitHub Actions*

Speculative Decoding for Language Models

[GitHub](#)

- Implemented and evaluated **novel decoding strategies** for causal language models in **PyTorch**, improving token prediction efficiency and reproducibility for open language model research.
- *Tools Used: PyTorch, Transformers, Python*

SecureVision: Raspberry Pi Biometric Authentication System

[GitHub](#)

- Developed an advanced **facial recognition security system** using Raspberry Pi, integrating computer vision techniques with **AWS** cloud services to capture, log, and alert on unauthorized access attempts, demonstrating expertise in IoT, machine learning, and cloud infrastructure.
- *Tools Used: Python, Boto3, AWS*

Language Model for Low-resource Text Data

[Report](#) [GitHub](#)

- Built n-gram language models (**unigram** & **bigram**) with **Add-k** and **Kneser-Ney smoothing**; evaluated using **perplexity** to measure linguistic predictability.
- *Tools Used: Python, NLTK, Pandas*

Sentiment Classification with Custom FFNN and RNN Architectures

[Report](#) [GitHub](#)

- Implemented sentiment analysis models from **scratch** using Feedforward Neural Networks (**FFNN**) with bag-of-words vectorization and Recurrent Neural Networks (**RNN**) with word embeddings.
- *Tools Used: Python, PyTorch, Numpy*

Graph-Based Heuristic Optimization

[GitHub](#)

- Modeled a test graph in C++ and applied a heuristic procedure to solve the **clique partitioning problem** (NP-hard) in dictionary-based test data compression for reducing data volume.
- *Tools Used: C++, Shell*

TinyML Drone

- Designed an **ESP32-CAM** based UAV with real-time edge AI, UART communication, CRC-8 error checking, and SLAM-based perception for plant health monitoring.
- *Tools Used: C++, ESP-IDF, Arduino, Espressify*

Speech Emotion Recognition - A Comprehensive Study

[Report](#) [GitHub](#)

- Built a **custom** neural network using PyTorch, achieving 99.5% test accuracy on the TESS dataset.
- Demonstrated improvements over traditional methods, such as Random Forest and SVM, by **2.5%**.
- *Tools Used: Python, Librosa*

Branch Prediction Analysis using GEM5

[Report 1](#) [Report 2](#) [GitHub](#)

- Evaluated branch predictors (LocalBP, TournamentBP) and cache settings on x86 architecture, using GEM5 for simulation. Created automated scripts in Bash to adjust source files dynamically and **optimize** configurations.
- *Tools Used: Gem5, Bash, Shell, CMake*

Publications

Total Citations: 140+ [Google Scholar](#)

- A hybrid meta-heuristic feature selection method for identification of Indian spoken languages from audio signals. (Das, A., Guha, S., et al.) **IEEE Access**. [Link](#)
- EmergeNet: A novel deep-learning based ensemble segmentation model for emergence timing detection of coleoptile. (Das, A., Das Choudhury, S., et al.) **Frontiers in Plant Science**. [Link](#)
- Hybrid feature selection method based on harmony search and naked mole-rat algorithms for spoken language identification from audio signals. (Guha, S., Das, A., et al.) **IEEE Access**. [Link](#)
- Flowerphenonet: Automated flower detection from multi-view image sequences using deep neural networks for temporal plant phenotyping analysis. (Das Choudhury, S., Guha, S., Das, A., et al.) **MDPI**. [Link](#)